

STAT 4530/6530

Homework 4

Due: Friday, March 27 via eLC by 11:59 pm

Formatting instructions: Please label everything clearly and make sure that it is organized. I recommend typing your solutions, but neatly handwritten solutions are acceptable. For problems that require statistical computing, please attach relevant R code and plots. **Please write down (directly below your name) whether you are enrolled in STAT 4530 or STAT 6530.**

Grading: Homework accounts for 40% of your final grade, and all homework assignments contribute equally to your final homework score. Your score for each homework assignment will be recorded as a percentage of the total points that you received out of the total possible points. The amount of points possible for each problem are indicated in parentheses.

Problems

1. (from [Gelman et al., pg. 37](#)) Placenta previa is an unusual pregnancy condition in which the placenta is implanted very low in the uterus, obstructing the fetus from a normal vaginal delivery. A study of the sex of placenta previa births in Germany found that there were 437 females among 980 placenta previa births. Let θ be the true proportion of female births in the placenta previa population.
 - (a) (2 points) What sampling model most appropriately describes Y , the number of females among the 980 placenta previa births from the Germany study?
 - (b) (4 points) Researcher A assumes there is no prior information about θ . Specify the appropriate prior distribution for θ and identify the posterior distribution (using derived results from class). Plot the prior and posterior distribution together in R.
 - (c) (4 points) Researcher B worked on a confidential pilot study, which saw 120 females out of 200 placenta previa births in Austria. Specify the appropriate prior distribution for θ and identify the posterior distribution (using derived results from class). Plot the prior and posterior distribution together in R.
 - (d) (4 points) For both Researchers A and B, use known formulas to find the posterior mean and variance. In each case, identify the posterior mean weights attributed to the prior and the observed data.
 - (e) (4 points) For both Researchers A and B, find a 95% percentile-based posterior credible interval for θ (in R).
 - (f) (4 points) How much evidence does this data provide for the claim that the proportion of female births in the placenta previa population is less than the proportion of female births in the general population, which is approximately 0.485? Answer this question by computing posterior probabilities (in R) for θ , using the posteriors for both Researchers A and B.

- (g) (4 points) Using all of the previous parts, summarize the above Bayesian analyses performed by Researchers A and B, being sure to comment on the influence of prior information on the posterior.
 - (h) (4 points) What might potentially make you weary about the prior used by Researcher B? Explain how you could adjust the prior in light of this.
2. Thus far, if we have binomial data, then we have explored using the prior distribution $\theta \sim \text{Beta}(a, b)$. The probability density function of this distribution is given by:

$$p(\theta) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \theta^{a-1} (1-\theta)^{b-1}, \quad \text{for } 0 < \theta < 1,$$

where,

$$\Gamma(z) = \int_0^\infty \theta^{z-1} e^{-\theta} d\theta,$$

is known as the **gamma function**.

- (a) (4 points) Show that a $\text{Beta}(1, 1)$ distribution is equivalent to a $\text{Unif}[0, 1]$ distribution, which has probability density function $p(\theta) = 1$ for $0 < \theta < 1$.
- (b) (4 points) Prove that for $\theta \sim \text{Beta}(a, b)$, the expected value is:

$$E(\theta) = \frac{a}{a+b}.$$

- (c) (4 points) In R, plot the beta distribution when $a = b = 1$, $a = b = 5$, and $a = b = 10$. Which of these hyperparameter settings would serve to provide the most prior information?
3. Suppose a licorice candy manufacturer produces pieces which are either red or black. Let θ be the true proportion of pieces which are red. You are tasked with estimating θ .
- (a) (4 points) You have done some investigative research, and have found that many believe this candy manufacturer packages licorice pieces in two different ways: individual packages are aimed to contain either 30% red, 70% black pieces, or 70% red, 30% black pieces. Describe what a prior distribution might look like which adequately describes this prior information. Why would the conjugate beta prior be inappropriate here?
 - (b) (4 points) It is challenging to encode the prior information from part (a) into a closed-form mathematical expression or distribution. However, we can still specify prior beliefs numerically. One way to do this in R is the following:

```
# Specify grid for values of theta from 0 to 1
grid = seq(0,1,length.out=1000)
# Constructing a prior which incorporates information from
  part (a)
prior_pdf = 0.5*dnorm(grid,mean=0.3,sd=0.05)+0.5*dnorm(grid,
  mean=0.7,sd=0.05)
prior_pdf = prior_pdf/sum(prior_pdf) # this step normalizes
  the prior pdf
```

Plot this prior distribution. What value would you change in the code above if you wanted to increase the “informativeness” of the prior? In your answer for this question, please attach a saved version of your plot, appropriately labeled.

- (c) (6 points) Let Y be the number of red licorice pieces out of a sample of size n . Assuming a binomial sampling model with the prior specified in part (b), use Bayes’ Theorem to write the posterior density $p(\theta|Y)$, up to a proportionality constant. Is this a recognizable closed-form distribution?

Note: We can express the prior distribution in part (b), up to a proportionality constant, as follows:

$$p(\theta) \propto \frac{1}{2}e^{-200(\theta-0.3)^2} + \frac{1}{2}e^{-200(\theta-0.7)^2}.$$

This is the form of a mixture of two normal distributions.

4. **This question is only required for students enrolled in STAT 6530.** (10 points)

Even though we do not get a recognizable closed-form distribution for the posterior in Question 3, part (c), we can still approximate and visualize the posterior using a technique called grid approximation. The idea here is similar to how the prior was specified in Question 3, part (b):

- Represent θ as a grid of 1000 equally-spaced values from 0 to 1.
- For each grid value of θ , we can compute the un-normalized posterior density (i.e, using the expression found in Question 3, part (c)).
- We normalize the values from the previous step by dividing by the sum of the densities over all of the grid values (this is an approximation of the integral used to compute the normalizing constant in Bayes’ Theorem).

The following R code replicates the process:

```
# Specify grid for values of theta from 0 to 1
grid = seq(0,1,length.out=1000)
# Compute the posterior pdf
prior_pdf = 0.5*dnorm(grid,mean=0.3,sd=0.05)+0.5*dnorm(grid,mean=0.7,sd=0.05)
bin_lik = dbinom(y,n,grid)
post_pdf = bin_lik*prior_pdf
post_pdf = post_pdf/sum(post_pdf) # this step normalizes the posterior pdf
```

Consider the following three sets of observed data

- Sample of 25 pieces, 13 of which are red
- Sample of 30 pieces, 15 of which are red
- Sample of 300 pieces, 150 of which are red

For each setting, use the above code to perform grid approximation of the posterior for θ (you will need to specify y and n before running the code). Plot each posterior, and summarize your observations in each scenario. In your answer for this question, please attach a saved version of your three plots, appropriately labeled.